

Algal Bloom Detection and Water Monitoring Using IOT Connectivity

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Abstract—This project is for an algal bloom detection using an internet of things system. It relies on microcontrollers, low-cost sensors, the cloud, machine learning and domain knowledge. The objective of this project is to detect the presence of harmful algae blooms using user defined thresholds. The purpose of this is to support environmental and water monitoring in lakes and rivers using low-cost systems. Through this paper, multiple machine learning algorithms and processes will be used to detect water quality and display this information in a cloud dashboard. This proof of concept is evidence that water quality monitoring can be done at local bodies of water at low-cost with good precision and accuracy results. This project highlights the need for more local water sampling and recording of sensor data in multiple bodies of water in British Columbia (BC) and how this can impact aquatic life, the environmental, and as well as access to safe recreational use of lakes and rivers.

Keywords—MQTT, IoT, Water Management, Algal Bloom, Machine Learning, ThingsBoard, Arduino

I. INTRODUCTION

While algae can be a natural and important part of the aquatic life cycle, it can start causing harm either when it is found in large amounts or when it is a certain algal species. “An algal bloom or algae bloom is a rapid increase or accumulation in the population of algae in fresh water or marine water systems” [1]. Algae is a broad term for photosynthetic, non-land plant organisms. Algal bloom detection is the process of monitoring environmental factors in the water and air and detecting when the levels of these organisms reach a level that poses a health risk for aquatic and human life. Algal bloom detection is an important aspect to water management and environmental risk management. The current process for water sampling and algae detection is a very labour-intensive process and requires close monitoring which can be expensive and time consuming. Traditional water testing tools and supplies can be costly and lack automation. This makes the domain a great opportunity to utilize an Internet of Things (IOT) system to support water health and public safety. Automatic water testing allows for live monitoring of bodies of water, and the use of machine learning allows the prediction of a harmful algal bloom before it can cause distress in the aquatic environment. This technology could be applied at low cost to multiple bodies of water around British Columbia. The objective of this project and report is to design and implement a predictive system that utilizes low-cost IOT sensors to measure freshwater quality indicators that can be used alongside machine learning to

predict the presence of harmful algae before it becomes a public health safety concern. Since there is no way to cost effectively measure algal growth directly, the system utilizes water quality measurements, such as temperature and pH, that can directly relate to and increase the risk of algal growth. These measurements are captured by sensors in the device and sent to an online ThingsBoard database, with a custom dashboard facilitating automation, monitoring, and remote control. From there, data can be exported for a machine learning model where it is analyzed and rated for risk based on a three-level scale: low, medium and high risk. This is shown in the machine learning model as algae present, algae may be present and algae is absent. The specific algal species marked for detection is cyanobacteria, and a data set available in the ABRI git repository is utilized. The system predicts levels of conductivity and total nitrogen, and it uses chlorophyll-a as a threshold. Overall, it can raise understanding of water quality in relation to algal blooms, and it combats the expensive cost of a chlorophyll sensor, as a main goal of this project is to remain cost-effective.

II. LITERATURE REVIEW

A. Jessika Broder

The paper titled “Chlorophyll Soft-Sensor Based on Machine Learning Models for Algal Bloom Predictions” discusses harmful algal blooms and how they pose a significant threat to aquatic ecosystems and public health. [3] The paper details how traditional water sampling and management is limited and laborious process. This aligns with what this project highlighted with a lack of readily available and local to BC datasets. This paper proposes a data driven approach that works with machine learning models integrated with automatic high-frequency monitoring using chlorophyll-a (Chl-a) soft-sensors. It focuses on three machine learning models: linear regression, classification and regression trees and random forests. The data collected included water temperature, pH, electrical conductivity, and battery voltage. These sensors served as cost-effective, energy efficient tools for real-time harmful algal bloom monitoring. The aspects retained from this paper include using low-cost sensors to track indicators of an algal bloom with additional sensors being added to improve metrics tracked. The sensors added include measuring phosphorus using RBG sensors and turbidity. Adding more indicators of water quality will help increase the accuracy of predicting harmful algal blooms. Many research papers still reply on manually sampling and

bridging the gap in real time monitoring is the goal of this IOT system being developed.

B. Casey Poppleton

Much like the previous paper, “Recent advances in algal bloom detection and prediction technology using machine learning” also talks about how harmful algal blooms can be predicted through machine learning (ML). [4] This paper focuses on 2 different research and technologies that are used for harmful algae blooms (HABs). It first goes over the research into different ML algorithms for algal bloom predictions (classification and regression). It talks about what models have been used with what inputs and what they specifically are trying to predict (target variables) and how they did. The next thing it goes over is how algae has been detected using image-based ML. It goes over what is sourcing the images, what algorithms are used, what species are being classified and how well it did. Lastly, it goes into the topic of explainable AI (XAI) focusing specifically on Shapley (SHAP) analysis looking at Chl-a concentration. This part of the paper mostly goes into explaining how using XAI they can pinpoint the variables that increase or decrease the concentration of Chl-a. We ended up using the information in the first part of this paper to help guide us through how to go about this process, specifically focusing on the classification part of it.

C. Aidan Evans

In the article, “Cloud-Based IoT System for Real-Time Harmful Algal Bloom Monitoring: Seamless ThingsBoard Integration via MQTT and REST API,” Annas et al. discuss how the IoT platform ThingsBoard can be used to gather, monitor, and organize data from IoT devices to predict and prevent Harmful Algal Bloom (HAB) threats. They primarily focus on explaining the communication protocols, cloud-platform setup, and modular ThingsBoard data visualization. In terms of connectivity, they found that the MQTT protocol was the most reliable for low-latency, real-time monitoring, while the REST API could handle irregular data uploads at the cost of higher latency [5]. This paper primarily influenced this project's choice to use the MQTT protocol with ThingsBoard. The integration between the device and the server ended up working extremely well, allowing the device to send and receive low-latency updates through the Docker-hosted ThingsBoard container.

D. Blain Polishak

In the article “Elevated pH Conditions Associated With Microcystis spp.”, Hampton et al. investigate the impact of harmful cyanobacterial algal blooms (CyanoHABs) on bodies of fresh water. They note that, during CyanoHABs, water pH often rises above 9.2, occasionally reaching as high as 11, due to the consumption of CO₂ during photosynthesis by cyanobacteria such as *Microcystis aeruginosa* [6]. Their findings not only indicate rising PH as a very favourable indicator of an active algal bloom, but also that high PH samples are optimal environments for the growth of cyanobacteria in general. They conclude that while *Microcystis* is largely unaffected by rising PH levels, the organisms with which it competes, such as diatoms, are

heavily suppressed by elevated PH levels. These findings suggest that monitoring pH can serve a dual purpose as it can act as an early warning indicator of conditions optimal for cyanobacterial proliferation, but also a direct indicator of an active CyanoHAB[6]. This paper helped guide the choice of sensors in the system design in addition to influencing the emphasis placed on measurements in the machine learning model, specifically pH.

E. Geordie

In “A review of the current and emerging detection methods of marine harmful micro algae” Abdul Rahman reviews the current detection and monitoring methods for algae, as well as how to prevent them. The paper categorizes existing detection methods into four main types: microscopic examination, cytochrome-based techniques, immunoassays, and nucleic acid-based detection. Each method has its strengths and weaknesses; for example, microscopy allows direct observation but is time-consuming, while DNA-based methods offer high accuracy but require advanced lab equipment (not applicable to the scope of our project). The review concludes that no single method is sufficient on its own; instead, integrating multiple techniques can improve detection accuracy and efficiency. The authors emphasize the need for developing faster, cost-effective, and automated detection systems to enable early warning and better management of HABs in marine environments.[7]

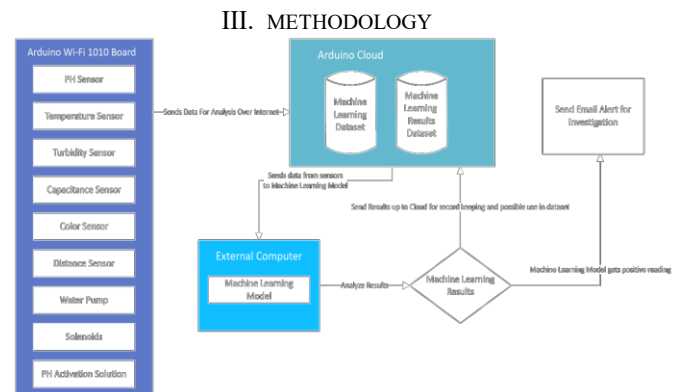


Fig 1. Initial Topology

The methodology of this project includes assembling a physical system for measuring environmental data, sending that data to a cloud interface, and using machine learning to find patterns to predict algal growth in water.

A. IOT Device and Componets

1) Required Componets

- Docker container running ThingsBoard
- Arduino MKR WIFI 1010
- Python (Machine Learning/API Calls)
- Algal Bloom Datasets
- GitHub
- Multiple IOT Sensors and Motors including:
 - I. PH (Acouto-B0D85DN3T4)
 - II. Water Temperature (3950-NTC)
 - III. Turbidity (TS-300B)

- IV. RGB (TCS-34725)
- V. Water Level Sensor (SL067)
- VI. Micro Pump (CJWP08-3.3V)
- VII. Solenoid (12V 6MM Inlet/Outlet)
- VIII. Buck/Boost Converter (XL6009)

2) System Design

The project design requirements dictated that the electronics should be away from the water's surface, to prevent accidental damage to its components. This requirement led to a system design which relies on small vacuum pumps to pull water samples into a separate testing reservoir, and which expels these samples using a normally closed solenoid attached to a drain. Research indicated that temperature, pH, dissolved phosphorus, and turbidity were key testing markers. Since no applicable sensors which measured dissolved phosphorus were available, a color-changing reagent combined with an RGB sensor was chosen to measure this concentration. A pH probe, waterproof thermistor probe, turbidity sensor, and water level sensor were also sourced, and after sourcing a reservoir, all components were modelled in Fusion-360 and virtually assembled into an organized system design.



Fig 1. Final System Design

B. Build Process

The system enclosure, including a custom lid to house sensors, breakout boards, the Arduino, and pumps, was designed in Fusion 360 and 3D printed. The turbidity sensor was mounted externally, through a hole in the side of the reservoir, and sealed in place with hot glue. The pH probe was mounted on a custom printed arm, and the temperature thermistor was slotted into a custom printed extension, to hold these sensors at their required heights for sample measurement. The color sensor was then adhered to the outside of the container, and the water level sensor was mounted in the lid, slightly above the other sensors, used to indicate when the reservoir reaches capacity. The reservoir

container was modified by drilling a drain hole at its bottom, to which a 6mm threaded hose attachment was secured. Horizontal holes were drilled at its base to ensure complete drainage, as the top of this drain sat above the bottom of the container. The components were then wired according to sensor datasheets and online diagrams, with the implementation of S8050 transistors to activate the vacuum pumps, which are wired to a separate 3.3V power input. An IRLZ44N MOSFET and XL6009 boost converter, set to 12V, were then connected to 3.3V power and solenoid, as the operating voltage of this solenoid is 12V.



Fig 2. Completed System Build

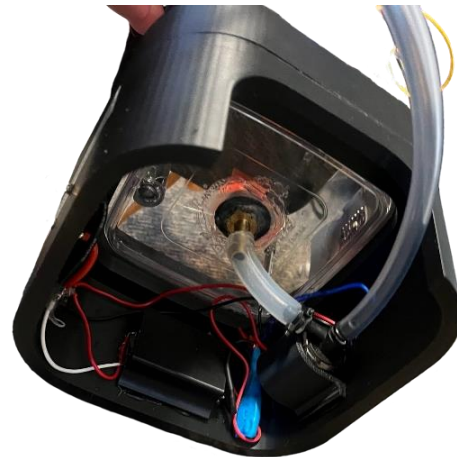


Fig 3. Lower Drainage System

C. Edge Logic

The term "Edge Logic" in this case encapsulates the local Arduino programming of the MKR1010 and how it handles failsafe detection, remote procedure calls, sensor data collection, and telemetry uploading. The backbone of the device is a timer that updates tasks periodically based on a user-defined duration. Separate timing intervals exist for failsafe polling and telemetry polling. Additionally, users can

define durations for individual actions to run. By default, failsafe detection polls frequently (every few seconds) while telemetry polls slowly (every few minutes). The structure of the primary loop is as follows: the first task that is checked is whether any failsafe has been triggered. This is to prioritize protecting the device before handling anything else. For instance, if the water sensor at the fill line has been submerged, the water pumping action would be interrupted to protect the electronics. Next, the device checks if the telemetry polling interval has passed. If so, data is collected from the sensors and aggregated into a JSON payload. To preserve the limited current of the MKR1010 and the power supply, sensors are polled one at a time. This is particularly important for components like the colour sensor which flashes a bright LED while taking measurements. After collecting data, the device immediately publishes an MQTT packet containing the information.

Outside of the core loop, the RPC requests are handled through an interrupt class that is always running parallel to the core loop. When an RPC call is received, the text is read and the appropriate task is run using the `startAction()` or `stopAction()` functions. These functions are generalized to handle any predetermined action easily, with safety measures built-in to ensure that only one action can be run at a time for the sake of both power conservation and reliability. For example, if a call is received with the text "startPumpCycle," then every other action is cancelled and the water pumping action is started. Actions themselves usually consist of raising or lowering the voltage of an output pin connected to a transistor that drives an actuator.

D. Cloud Setup

The cloud functionality can be separated into three major components: Docker, ThingsBoard, and the MQTT and RPC protocols. The online server was run using Docker, which is an application that allows computers to host application containers. In this case, the server was hosted on a Windows PC running a Docker container. In the future, it would be more efficient to either host the server on a dedicated Linux device like a Raspberry Pi or use an online service like Firebase to host the server. Through Docker, the Community Edition of ThingsBoard is run as a container exposed to the public internet. ThingsBoard is an open-source self-hosted web service that allows users to create and manage IoT devices. It has the added benefits of using a PostgreSQL database, as well as having built-in tools for dashboard creation, automation, and customization. This made the process of developing the online system simple and adjustable for future improvements.



Fig. 4 ThingsBoard Dashboard

The dashboard contains action buttons, live status updates, and a recent telemetry graph. Using the interface, any action can be started or cancelled at any time. The device is programmed such that anything can be interrupted for safety reasons. For communication with the online server, the MQTT and RPC protocols were used. MQTT is a publish/subscribe protocol designed for IoT devices, allowing them to send data with minimal overhead. In this case, it is used to upload JSON payloads containing telemetry data from the device to the server. Then, RPC is used by the server to send direct procedure requests to the device. This facilitates low-latency action calls for starting the pump, pulling reagent solution, stopping actions, and more. This facilitates low-latency action calls.

E. Dataset

There is a lack of available and usable datasets pertaining to algal sensor data. The dataset used in this project was an unlabeled dataset with water samples from unknown locations. The feature set included in the dataset was temperature, pH, turbidity, total phosphorus, total nitrogen, conductivity, source being river or lake, and other sensor data. The dataset was a part of a GitHub repository linked below [2]. This linked project used both sensor data as well as algal satellite image processing. The goal of the project was to create an inexpensive way of detecting algal growth, so only inexpensive sensors were used. These included temperature, turbidity, RGB values for detecting total phosphorus, and pH.

F. Machine Learning for Detection

1) Data preprocessing

To ensure the data is ready for multiple machine learning algorithms some preprocessing of the data was done. To start, we removed any unnecessary columns from the dataset. The columns that we used varied throughout the project, however, the columns used to predict an algal bloom were pH, Turbidity, Temperature and Total Phosphorus (TP).

After removing those columns, we then had to further process the data to reduce the dimensions of it for proper visualization and help with prediction. To do this we used Principle Component Analysis (PCA). This reduced our 4 columns into 2 Principle Components which simplified the data while retaining the most important information. The most important information is gathered by taking the highest

combined variance of all 4 columns and putting that into the first Principle Component and then the next highest variance for the next Principle Component [21].

Lastly, we performed a Train Test Split on the dataset so we could accurately measure our precision, recall and f1-score. After processing the data, we could then move on to Machine Learning.

2) Unsupervised

Since there was an unlabeled dataset used, it was logical to start with unsupervised machine learning algorithms. Unsupervised learning is when you use unlabeled data or non-predefined categories to find relationships and patterns in data [8]. This led to the use of K-Means clustering [9]. This algorithm groups similar data into clusters.

When we actually went to test K-Means Clustering, we first needed to determine how many clusters we wanted to generate. To do this, we employed the use of the Elbow Method and Silhouette Analysis. What these do is help us determine the optimal number of clusters to use when generating the labels using K-Means Clustering. They determine this using slightly differing methods which when combined, give us a more concrete result to use.

The Elbow Method determines this by plotting the Within-Cluster Sum of Squares (WCSS) against the number of clusters (k) and looking for the point where improvement of WCSS doesn't significantly change with an increase of k (The Elbow). WCSS shows how close the data points are to the center of their respective cluster [10].

Silhouette Analysis does this through plotting the Silhouette Score against K and determining the k where the score is the highest. Silhouette Score ranges from negative one to positive one and determines if the points in the clusters have good cohesion and the clusters are well separated [11].

The results we got from these two methods are shown in Fig 6. Below:

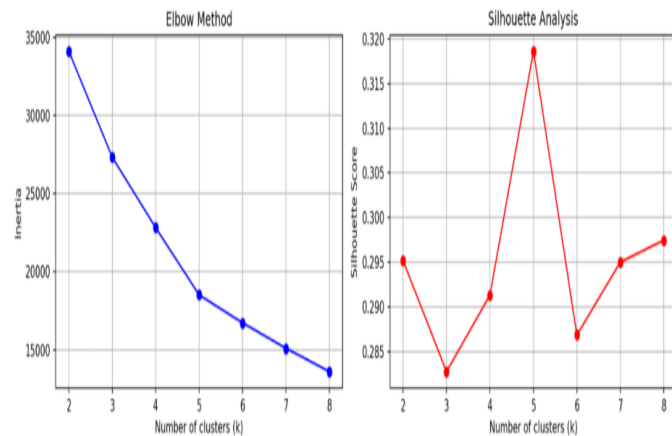


Fig 5 Elbow Method and Silhouette Score

3) Semi Supervised Pseudo Labelling

After the results of K-means clustering, it was decided to use the clustered created as pseudo-labels [12] for semi-supervised learning using Random Forest (RF) and Gradient Boosting (GB) algorithms. We decided to choose these algorithms as we found several studies that used different tree-based algorithms such as “Machine learning-based prediction of harmful algal blooms in water supply reservoirs” where they used Extreme Gradient Boosting

(XGB) and RF. They found that these algorithms both had good prediction performances, but that XGB was more stable [13].

4) Supervised with Surrogate Target Threshold

Based on the unreliable results of semi supervised learning, threshold levels were set to create a target for supervised learning. These thresholds relied on domain knowledge and research to be applicable to lakes and rivers around BC. [14] The threshold values decided on were based on the three-level scale mentioned above. We choose to use conductivity and total nitrogen as the targets for these levels due to them not being any of the sensors we are currently using. These created proxy labels. [15]

```
#set target for algae presence based on conductivity AND TN
def classify_bloom(conductivity, TN):
    if conductivity > 150 and TN > 0.1:
        return 'algae_present'
    elif (100 < conductivity <= 150) and (0.05 < TN <= 0.1):
        return 'algae_may_be_present'
    else:
        return 'algae_absent'
```

Fig 6 Surrogate Threshold Values

We decided to use target features that were currently not being measured in the field since when we used features from the sensors we have, the results were artificially high. Including target values within the feature set leads to models appearing to perform well during training but fail drastically in real-world applications. Using a correlation matrix showing correlation, we decided to use the two values that had the strongest correlation while still being relatively inexpensive sensors that could be added to verify the results if needed.

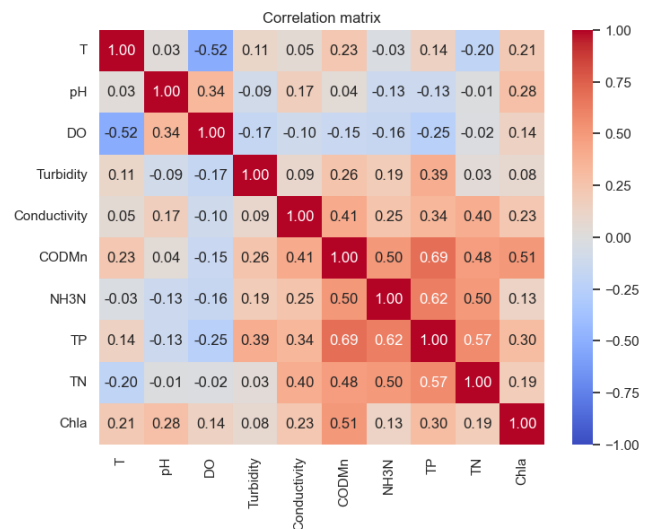


Fig 7 Correlation Matrix

This led to using a two staged approach that combined a supervised model and related the predicted values to algal growth risk using domain knowledge. Total nitrogen was used as it can have a fairly large impact that creates high risk conditions for algal blooms. The typical values for total nitrogen are 1 -2 mg/l concentration for high risk. [16] We

decided to use 0.1 as the cutoff for high risk to ensure that the system and model contain high precision and does not miss an indication of algal growth. Conductivity was added to create a proxy-indicator as it can be very site and location specific. [17] Typically, conductivity values of 0 – 200 are natural, low-nutrient systems. Anything above 300 often indicates nutrient enrichment that can promote algal growth. We decided to 150 as the cutoff for conductivity so it had a higher chance of detecting the level and therefore not missing any indicators of algal growth. With more time and domain expertise, these values could be better refined to match the environmental climate in BC. We also experimented using chlorophyll-a (Chla) as the surrogate threshold value as this is an actual risk of bloom indicator used in BC. These are very expensive sensors, but these values are only used for training the model. Based on this, the model is trying to predict Chla levels using sensor data we can gather with inexpensive sensors. Doing this gave us very accurate results that made sense with the domain knowledge found.

```
def classify_bloom(Chla):
    if Chla >= 33:
        return 'algae_present'
    elif 33 > Chla >= 10:
        return 'algae_may_be_present'
    else:
        return 'algae_absent'
```

Fig 8 Chla Threshold Values

These values were taken from ‘Guidelines for Canadian Recreational Water Quality – Cyanobacteria and their Toxins’. [18] It stated in section 1.2 that the guideline values for primary contact recreation are total chlorophyll-a 33 µg per liter.

G. Alerting User

This alert is set up through a Gmail Account that is being accessed by the EmailMessage Library through an app password. This alert only gets sent if there is at least 1 item in the `algae_present` class.

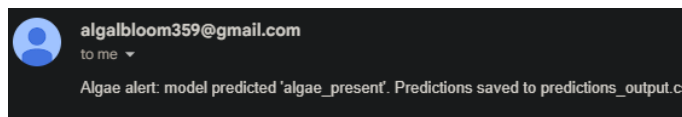


Fig 10 Email Alert

IV. RESULTS AND DISCUSSION

A. Evaluation Metrics

Interpreting machine learning results includes understanding the difference between evaluation metrics such as accuracy, precision, recall, and an F1 score. Accuracy is the basic metric for evaluation and is just an average of the results. Precision was the metric focused on this project, and

it includes fewer false positives. That means that the system is more specific and generates fewer incorrect positive results or a “false alarm”. Recall includes fewer false negatives, and this means the system is more sensitive. This is important where missing a case is dangerous. [19]

		Predicted	
		Animal	Not animal
Actual	Animal	True Positives	False Negatives
	Not animal	False Positives	True Negatives

Fig 11 Evaluation Metrics

B. Unsupervised

Using the previously mentioned Elbow Method and Silhouette Analysis, we determined that the number of clusters needed was 5. After determining that, we ran the K-Means Clustering algorithm on our data and received the following graph in Fig. 12

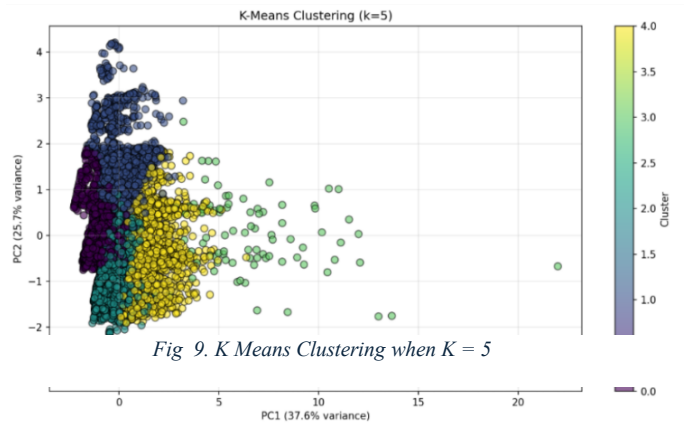


Fig 9. K Means Clustering when K = 5

After examining the K-Means Clustering graph and using our knowledge of what our data meant in terms of what is and isn't and algal bloom, we determined that the further right the points were, the more of a chance there was that it would be an algal bloom. We also determined that while there were clear clusters that the algorithm determined, those clusters really could be anywhere and would look correct. To confirm our suspicions, we attempted to classify the testing data using

K-Means Clustering and received very poor results as shown in Fig. 13

```
Predictions saved to predictions_output.csv
Total predictions: 2740

Number of unique clusters found: 1
Cluster distribution: [ 0  0 2740]
```

Fig 12 Testing Run with K Means Clustering

Because the algorithm only classified all the testing data into one cluster, we decided that this was not the correct method to properly classify the data that we had.

C. Semi Supervised

After running the RF and GB algorithms, Fig. 14 and Fig 15 show the unstable and poor results of the RF and GB classification report.

```
Random Forest classification report (vs K-Means):
      precision    recall  f1-score   support

0         0.0000    0.0000    0.0000         0
1         0.0000    0.0000    0.0000         0
2         1.0000    0.6507    0.7884       2740
3         0.0000    0.0000    0.0000         0
4         0.0000    0.0000    0.0000         0

accuracy          0.6507       2740
macro avg         0.2000    0.1301    0.1577       2740
weighted avg      1.0000    0.6507    0.7884       2740
```

Fig 13 Random Forest Classification report using K Means

```
Gradient Boosting classification report (vs K-Means):
      precision    recall  f1-score   support

0         0.0000    0.0000    0.0000         0
1         0.0000    0.0000    0.0000         0
2         1.0000    0.6507    0.7884       2740
3         0.0000    0.0000    0.0000         0
4         0.0000    0.0000    0.0000         0

accuracy          0.6507       2740
macro avg         0.2000    0.1301    0.1577       2740
weighted avg      1.0000    0.6507    0.7884       2740
```

Fig 14 Gradient Boosting Classification Result using K Means

Due to the very poor results received, we decided to move on from this approach as this didn't seem to be the appropriate method for our dataset.

D. Supervised with Surrogate Target Threshold

The last approach we used was supervised learning with surrogate target thresholds. The results of these surrogate targets were much more reliable and made more sense with domain knowledge. We decided to keep with the same random forest and gradient boost algorithm as there was previous research that had good results with those robust and adaptive algorithms. The first set of thresholds that were used were total nitrogen and conductivity. The results are shown on Fig. 16 below:

1) Nitrogen and Conductivity

Unfortunately, during testing, both the random forest and gradient boosting algorithms could not predict the middle range representing that algae may be present. This was mostly because the model predicted high nitrogen and conductivity, but these factors can be very dependent on total phosphorus content and many other complex factors. The model was only able to detect high ranges and struggled with mid and lower ranges of these features during training.

```
Random Forest:
      precision    recall  f1-score   support

algae_absent      0.79      0.68      0.73         774
algae_may_be_present  0.00      0.00      0.00          0
algae_present      0.88      0.93      0.90       1966

accuracy          0.86       2740
macro avg         0.56      0.54      0.55       2740
weighted avg      0.86      0.86      0.86       2740

Gradient Boosting:
      precision    recall  f1-score   support

algae_absent      0.79      0.51      0.62         774
algae_may_be_present  0.00      0.00      0.00          0
algae_present      0.83      0.95      0.88       1966

accuracy          0.82       2740
macro avg         0.54      0.48      0.50       2740
weighted avg      0.82      0.82      0.81       2740
```

Fig 15 Random Forest and Gradient Boosting Training Model Results

After seeing the results of the training, we still decided to run it on some testing data the model had not seen. This is shown on Fig. 17 below. As expected, the model could not predict the middle range. Both models had a high agreement rate.

```
PREDICTION DISTRIBUTIONS:
=====
K-Means cluster distribution:
2      2740
Name: count, dtype: int64

Random Forest predictions distribution:
algae_present      2070
algae_absent         670
Name: count, dtype: int64

Gradient Boosting predictions distribution:
algae_present      2223
algae_absent         517
Name: count, dtype: int64

=====
MODEL AGREEMENT (RF vs GB):
=====
Agreement rate: 2451/2740 (89.45%)

Random Forest vs Gradient Boosting comparison:
GB      algae_absent  algae_present  All
RF
algae_absent      449          221      670
algae_present      68          2002     2070
All                517          2223     2740
```

Fig 16 Prediction Distribution Using Nitrogen and Conductivity

Since the results were still not at the level we needed and that the model could not predict middle ranges that matched the three-level risk used for algae growth. This led to the exploration of using chlorophyll-a values to train models.

2) Chlorophyll-A

Using chlorophyll-a (Chla) values to train both the random forest and gradient boosting models created much more balanced results. Since this project does not use chlorophyll-a values in predicting (feature set) we were able to use it for training the model as surrogate target values. Chlorophyll-a (Chla) sensors are incredibly expensive but since it is a laboratory equipment and measures the actual count of chlorophyll-a in a water sample, it is reliable and could be used for training only. There was no need to verify the output as these sensors are regularly monitored and calibrated. Using the threshold values shown in Fig. 9 we were able to get the training results shown in Fig. 18 below.

Random Forest:				
	precision	recall	f1-score	support
algae_absent	0.83	0.90	0.87	1841
algae_may_be_present	0.63	0.53	0.58	709
algae_present	0.69	0.52	0.59	190
accuracy			0.78	2740
macro avg	0.72	0.65	0.68	2740
weighted avg	0.77	0.78	0.77	2740
Gradient Boosting:				
	precision	recall	f1-score	support
algae_absent	0.78	0.92	0.85	1841
algae_may_be_present	0.56	0.38	0.45	709
algae_present	0.75	0.41	0.53	190
accuracy			0.74	2740
macro avg	0.70	0.57	0.61	2740
weighted avg	0.72	0.74	0.72	2740

Fig. 17. RF and GB results for training using Chla threshold

When these models were used on testing data, it was able to predict all three categories and the models had good agreement when classifying as shown in Fig. 19.

Predictions saved to predictions_output.csv				
Random Forest predictions distribution:				
algae_absent	2027			
algae_may_be_present	572			
algae_present	141			
Name: count, dtype: int64				
Gradient Boosting predictions distribution:				
algae_absent	2180			
algae_may_be_present	460			
algae_present	100			
Name: count, dtype: int64				
=====				
MODEL AGREEMENT (RF vs GB):				
=====				
Agreement rate: 2322/2740 (84.74%)				
Random Forest vs Gradient Boosting comparison:				
GB	algae_absent	algae_may_be_present	algae_present	All
RF				
algae_absent	1917	102	8	2027
algae_may_be_present	245	320	7	572
algae_present	18	38	85	141
All	2180	460	100	2740

Fig. 18 Prediction Distribution for Test Data Using Chla Threshold

E. Results using IOT System Data

We ran these models on sensor data collected from the IOT system and got the results shown on Fig. 20. The model's agreement rate was low, but the gradient boosting displayed expected results for the data given. It was able to predict the three levels of algal bloom risk.

Random Forest predictions distribution:				
algae_absent	998			
algae_may_be_present	49			
algae_present	10			
Name: count, dtype: int64				
Gradient Boosting predictions distribution:				
algae_may_be_present	738			
algae_absent	194			
algae_present	125			
Name: count, dtype: int64				
=====				
MODEL AGREEMENT (RF vs GB):				
=====				
Agreement rate: 215/1057 (20.34%)				
Random Forest vs Gradient Boosting comparison:				
GB	algae_absent	algae_may_be_present	algae_present	All
RF				
algae_absent	194	727	77	998
algae_may_be_present	0	11	38	49
algae_present	0	0	10	10
All	194	738	125	1057

Fig. 19 RF and GB Results Using IOT System Data

F. Machine Learning Constraints

The main constraint that we ran into with machine learning was our dataset. One aspect of this was the fact that the data was unlabeled. This vastly increased our challenge with performing machine learning as we were unsure where to start. This also slowed us down as we had to try multiple different methods (discussed above) before getting something that succeeded. The lack of labels also not knowing if any of the rows within it were from an algal bloom. If we had more time, we could have collected a dataset and labelled it for our needs. The other option is we could have contacted someone to receive a better dataset than the one we used. The next constraint that we faced was not having a physical source of algae to test our system with. If we were doing this project closer to when cyanobacteria were in bloom, we could have collected multiple water samples to test our final Machine Learning Algorithm against. The last constraint is that we didn't have the materials to perform the Molybdenum blue phosphorus method [20]. If given more time, we could have connected with other professors in the science department to gain access to the materials necessary to properly measure the Total Phosphorus in the water, rather than using blue food dye.

V. CONCLUSIONS AND RECOMMENDATIONS

Through exploring water quality analysis and algae detection, there is a large lack of available data and low-cost water management systems. Using our inexpensive water quality sensor IOT project, we were able to help predict the risk of algal bloom using water quality health indicators. Through the time spent researching and exploring the data, the recommendations this project has brought forward are shown below:

A. System

Although functional as a demonstration unit, the system would require extensive redesign with an emphasis on optimization to be practical for field deployment. While core functionality was successfully demonstrated, there are several prominent drawbacks to the hardware implemented in the current design, leading to inefficiency and

impracticality. Most notably, the size of the reservoir is far too large for practical field applications, leading to substantial wasted time and energy when pulling and draining samples. Furthermore, the large testing volume requires the use of excessive reagent, leading to environmental concerns, and would theoretically require the system to be serviced and refilled far too often. Subsequently, despite using a 3.3V power input, allowing the system to be powered by industry-standardized Li-ion batteries, the use of a 12V solenoid to drain the system leads to substantial power loss. Although the XL6009 boost converter runs with an expected efficiency of 80% when actively converting 3.3V to 12V, in a field deployment this system would spend the majority of the time inactive. With an expected 30mA no-load current, based on its datasheet, this converter would drain most applicable Li-ion batteries in mere days, and is far too inefficient to be implemented in the field. Finally, the limited budget of this project resulted in a cheap pH measurement probe being utilized in this system. This probe is extremely volatile and requires frequent and intricate maintenance to produce reliable data. Consequently, in a revised system, an alternative and more robust method of pH measurement would be required, or pH would need to be entirely omitted as a marker. If this system were to be redesigned for a practical implementation, the use of a much smaller reservoir would also be a necessity. Reducing reservoir size would substantially decrease the duration of sample collection, in turn reducing the power drawn during sample intake and drainage. It would also reduce the volume of reagent required to test a sample, minimizing the system's environmental impact and extending the required intervals to service and refill the system. Finally, the current drainage system would need to be replaced with a low power solenoid that could run directly on 3.3V power from a Li-ion battery, circumventing the need for a boost converter and the associated passive power consumption.

B. Machine Learning

To increase the results of the machine learning algorithms and allow it to adapt to the changing water environments around BC, a labeled dataset would be needed to continue this project. Domain and expert knowledge would also assist this project in getting to the next step and increasing the reliability of the results. Going forward, we recommend spending more time gathering data, exploring the data and the use of different algorithms and expanding domain knowledge.

Please see reference [22] for GitHub link.

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